

AP CSP | Lesson 04: Test & Improve

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 40 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Chapter 1 Check: 12 original questions, 30 classroom marks, about 40 minutes. Q5 requires exactly two selections. Show traces, calculations and reasons; retain your Ticket Total code and development record.

1. A ticket scanner stores the eight-bit nonnegative binary integer 00101101. What decimal value does this represent? [1 mark]

A. 43 B. 45 C. 46 D. 53

Answer: _____

2. A ticket kiosk receives a visitor's selection of two tickets, calculates a total, and shows 27 on its screen. Which statement correctly identifies input and output? [1 mark]

A. Input: the total calculation; output: the visitor's selection. B. Input: the screen; output: the stored price variable.
C. Input: the visitor's selection; output: the displayed 27. D. Input: the displayed 27; output: the multiplication operation.

Answer: _____

3. Consider the following AP-style pseudocode. [1 mark]

$x \leftarrow 4$

$y \leftarrow x + 2$

$x \leftarrow 9$

DISPLAY(y)

What is displayed?

A. 4 B. 6 C. 9 D. 11

Answer: _____

4. In situation I, a parser rejects DISPLAY(total because a required closing parenthesis is missing. In situation II, a language reports division by zero as an error while executing an otherwise valid statement. Which classification is correct? [1 mark]

A. I: logic; II: syntax B. I: run-time; II: logic C. I: syntax; II: logic D. I: syntax; II: run-time

Answer: _____

5. Ticket Total accepts integer inputs from 1 to 50. Each ticket costs 12, plus a fee of 3 once per order. A faulty version computes tickets * (12 + 3). Which TWO valid inputs expose this error? Select exactly TWO answers. [1 mark]

A. 2 B. 1 C. 50 D. 0

Answer: _____

6. A visitor says, "The displayed 27 does not tell me whether the booking fee is included." Which response best demonstrates iterative development? [1 mark]

A. Revise the output label to address the feedback, then check the visitor's understanding and rerun the numeric tests. B. Keep the output unchanged because the program has no syntax errors.
C. Add another fee of 3 to every result so that the visitor notices it. D. Ask only the original developer whether the label is clear and stop testing.

Answer: _____

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Written responses | Questions 7-8

7. [4 marks]

A museum needs distinct fixed-length binary identifiers for 41 exhibits. No patterns are reserved.

(a) Find the minimum number of bits per identifier. Show both why it is sufficient and why one fewer bit is insufficient.

(b) With that length, how many different patterns exist?

(c) If the patterns represent consecutive nonnegative integers starting at 0, what is the largest integer?

8. [4 marks]

A translation app records a visitor's spoken question, uses a program to produce a translation, and displays translated text.

(a) Explain why this is a computing innovation.

(b) Identify one input to the program and one output from it.

(c) A visitor says that the displayed text is too small to read. Suggest a change and explain how you would check whether it helps.

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Written responses | Questions 9-10

9.

[4 marks]

Consider the following AP-style pseudocode.

$a \leftarrow 4$

$b \leftarrow a + 3$

$a \leftarrow 10$

$b \leftarrow b + 2$

DISPLAY(a)

DISPLAY(b)

(a) State the values of a and b after the second statement.

(b) State their values after the fourth statement.

(c) Give the two displayed values in order.

(d) Explain why changing a to 10 does not automatically change b.

10.

[4 marks]

A program should calculate a rectangle's perimeter from positive integer length and width inputs.

length $\leftarrow$ INPUT()

width $\leftarrow$ INPUT()

perimeter $\leftarrow$ length + width

DISPLAY(perimeter)

(a) Replace the incorrect assignment with a correct statement.

(b) For inputs 7 and 4 in that order, give the corrected output.

(c) Explain why the original statement is a logic error, assuming its syntax is valid.

(d) Describe the purpose of the corrected program, referring to its inputs and output.

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Written responses | Questions 11-12

11. [4 marks]

Ticket Total assumes valid integer inputs from 1 to 50. Tickets cost 12 each, plus 3 once per order. A faulty program computes $\text{total} \leftarrow \text{tickets} * (12 + 3)$.

- (a) Give the correct expected totals for valid boundary inputs 1 and 50.
- (b) For input 2, give the expected total and the faulty program's actual total.
- (c) Explain the cause of the mismatch.
- (d) Write a corrected assignment for total.

12. [4 marks]

Two students use pair programming to develop Ticket Total. Its calculation is correct, but a visitor cannot tell whether the displayed total includes the booking fee. One student adapted the input lines from the teacher's starter code.

- (a) State one specific navigator action that contributes to the work.
- (b) Suggest a revision based on the visitor's feedback.
- (c) State what should be checked after that revision.
- (d) Write a suitable acknowledgement for the adapted code.
